Task #1
Dataset Understanding and Chapter 2: Probability Applications

Task Objective

Understand your selected dataset, prepare it for analysis, and apply the probability rules and concepts covered in Chapter 2 to answer meaningful questions and extract useful insights.

This task is the first step toward your final project. In the following tasks, you will apply concepts from other chapters to the same dataset, gradually building your analysis for the final project.

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1. 1. Dataset Requirements

- You may modify or replace the dataset you previously selected.
- Your dataset must have potential for use in a relevant *conference, hackathon, or competition*.
- After Task 1, you must continue using the same dataset for the remaining tasks and the final project.

1.2 Dataset Understanding and Preparation

Your analysis must begin with a clear description of your dataset.

- Report the number of files, rows, and columns ,…etc.
- Describe the variables, their meanings, and their data types (numerical, categorical, text, date/time, etc.).
- Explain what each row represents and how the files are related.
- Show a small sample of the data.
- Identify relevant missing values, duplicate records, and data limitations.
- If your dataset is not tabular, transform it into a suitable tabular format for statistical analysis and explain the transformation.
- Document all preprocessing steps, including cleaning, filtering, removing rows, handling missing values, removing duplicates, merging files, and other transformations, where applicable.

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2. Chapter 2: Probability Applications

Apply most of the relevant probability rules and concepts covered in Chapter 2, including sample spaces and events, counting techniques, probability axioms, addition and multiplication rules, conditional probability, total probability, independence, and Bayes' theorem.

For each application:

1. Define a meaningful question related to your dataset.
2. Identify the relevant events and assumptions.
3. Select and explain the appropriate rule.
4. Show the calculation and present the result.
5. Interpret the result in the context of your data.

Do not apply a rule simply to mention it. Explain its relevance to your analysis. If a concept cannot be applied directly, explain why and use a suitable alternative.

3. Results Validation

Validate your major results using the dataset itself.

- Compare results calculated using probability rules with results obtained through direct calculations from the relevant observations.
- Use independent calculation approaches where possible, even if both are implemented in Python.
- Explain why the results agree or investigate any differences.
- Clearly state any assumptions and distinguish between exact agreement and differences caused by sampling or modeling assumptions.

4. Insights and Data Presentation

- Use suitable probability tables, charts, or visualizations to summarize your findings.
- Explain the main patterns, relationships, or differences revealed by your analysis.
- Discuss the assumptions and limitations of your conclusions.
- Explain how your findings could be useful in the proposed conference, hackathon, or competition.

Python and Google Colab Requirements

Python is the only programming language permitted.

Submit a Google Colab notebook containing the complete code and explanations for your analysis.

The notebook must:

- Start with a text cell explaining the problem, objectives, and analysis steps.
- Be organized into clear sections, with text cells explaining each major step before the corresponding code.
- Include well-commented, readable, and logically organized code.
- Include all data loading, preprocessing, statistical calculations, validation, and visualizations.
- Explain the statistical rules used and how they are applied to the dataset.
- Present the results and interpret their meaning.
- Be reproducible, so *the analysis can be rerun and reused in later tasks.*

Data preprocessing and reproducibility:

- Clearly document every major preprocessing step and explain why it is needed.
- Report the number of rows before and after each major filtering or removal step.
- Justify decisions to remove, filter, or transform observations.
- Explain how preprocessing decisions may affect the calculated probabilities and conclusions.
- *Preserve the original data and perform preprocessing through documented code whenever possible, so the same steps can be reused in future tasks and the final project*.

Presentation Requirements

- Prepare a PowerPoint presentation of **up to 15 slides**.
- The presentation must not exceed **8 minutes**.
- Slide 1 must describe the dataset structure, including the number of files, rows, and columns, variable types, and data content, where applicable.
- Present the problem, statistical applications, calculations, validation, findings, and conclusions clearly and concisely.
- Only one team member will present each task.

- Every team member must present at least one task throughout the course. The presenting student must be different for each task until all team members have presented.
- All team members must understand the entire task and be ready to answer questions during the discussion, regardless of who presents.

Submission

Submit the following two items:

1. **Presentation:** Export your slides as a PDF file.
2. **Google Colab notebook:** (Submit the link to your Google Colab notebook) Include the complete code, explanations, calculations, validation, and visualizations.

No separate written report or additional files are required. *Ensure the instructor can access your notebook.*

Competition and Assessment

Teams will be evaluated based on:

- Understanding of the dataset and its structure.
- Quality of data preparation and justification of preprocessing decisions.
- Coverage and correct application of **Chapter 2** concepts.
- Reliability and validation of the results.
- Creativity in formulating questions and selecting statistical methods.
- Quality of insights and their interpretation.
- Code organization, readability, documentation, and reproducibility.
- Clarity of the presentation.
- Individual understanding during the discussion and Q&A.

Creativity is encouraged in the way you explore the data, formulate questions, organize your code, select appropriate probability concepts, and apply them to the dataset.

Remember: These tasks are designed to build on one another. The dataset, preprocessing steps, and code developed in Task 1 should provide a foundation for the remaining tasks and the final project.